

Effects of Verbal Interruption in Conversations with an Intelligent Virtual Agent in Virtual Reality

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Abstract

Intelligent assistants powered by large-language models (LLMs) are increasingly becoming a part of our daily lives, particularly on our mobile devices. In immersive virtual reality (VR), intelligent assistants can be embodied as intelligent virtual agents (IVAs). They are commonly humanoid characters capable of multimodal interaction with the user. When prompted with a question, these assistants or IVAs typically formulate elaborate responses, which are then vocalized. While it is possible to terminate the output by tapping on a mobile phone screen, in immersive VR, different strategies have to be investigated to interrupt the IVA. In our work, we investigated whether implementing the possibility to interrupt embodied IVAs in VR enhances the user experience and the users' perception of the IVA. Therefore, we conducted a user study in which 30 participants were tasked to answer several quiz questions with the help of an IVA, with or without the ability to interrupt. The results show that the possibility of interrupting the IVA increases perceived efficiency, attractiveness and stimulation. Qualitative feedback suggests that real-time interruptions lead to more natural and dynamic conversations. Although the interruption possibility did not change users' perception of the IVA regarding perceived anthropomorphism, likeability, and perceived intelligence, we discussed these findings along with participants' qualitative feedback and provide various design insights for future research.

CCS Concepts

• **Human-centered computing** → **Virtual reality**.

Keywords

Conversing with virtual agent, Interrupting an AI, Virtual Reality, Human-agent-conversation, Turn-taking, Verbal interruption, Voice-based conversations, Intelligent Virtual Agents

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1 Introduction

Artificial intelligence (AI) based on large language models (LLM) is becoming increasingly widespread in everyday life. These systems have the capacity to comprehend and generate texts, thus assisting individuals in a wide range of applications. While some individuals seek answers in written format, others engage in verbal interactions with the AI.

In virtual reality (VR), intelligent virtual agents (IVAs) can facilitate the human-AI interaction by giving AI a human-like appearance and behavior [9]. It has been shown that among different levels of anthropomorphism (e.g., from voice-only to full-body representation), a full-body representation of humanoid virtual agents leads to the highest degree of perceived social presence [11, 16, 22]. Social and co-presence is an illusion that can occur in VR where users feel that computer-generated virtual humans or characters are with them in the same environment and their actions are actually happening in the real world [28]. This means that, for example,



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IVAs' verbal and non-verbal communication cues such as smiling or nodding the head in a conversation with the user are perceived as if they were real [20]. This can be then used to implement more natural human-AI conversations through IVAs.

Conversations with AI are currently associated with some limitations. For example, produced responses are often extensive, even when a concise answer would suffice. In text-based interactions, users can simply press a stop button to interrupt the output or skim through it. However, in VR, where keyboards are less efficient and interaction is primarily based on speech or 3D user interfaces (UIs), interruption through conventional methods is less practical. Unlike text-based UIs, where stopping output with a button feels natural, using a similar mechanism may feel disruptive and unintuitive when the interaction with the AI is based on speech with an anthropomorphic agent. Therefore, speech-based interactions in VR should allow for more natural ways to manage conversation flow, such as verbal interruptions or adaptive response mechanisms.

Conversations between humans are characterized by great dynamism. Humans are typically capable of managing intuitive turn-taking cues [25], particularly through the use of interruptions or brief pauses. Researchers have also started analyzing human-IVA conversations to integrate interrupting capabilities, similar to human-human conversations [31, 32].

However, these early attempts lack an empirical evaluation of the work which motivated us to follow up on this topic to i) enable users to interrupt an IVA in VR and ii) empirically evaluate its effects on their experience with and their perception of the IVA. In order to conduct research on this topic, it is necessary to consider the following research question:

How does the ability to verbally interrupt an IVA in VR influence users' experience and perception of the IVA?

We defined our hypotheses as follows:

- **H1:** If the user can verbally interrupt the IVA's flow of speech and direct the conversation, the user experience is improved.
- **H2:** The ability to interrupt the AI will make the conversation seem more natural.

2 Related Work

Natural conversations and turn-taking management. In the context of verbal communication between individuals, it is generally considered optimal for one person to assume the role of speaker while the other assumes the role of listener. Once the speaker has finished talking, the roles are then reversed. This approach allows for an exchange of information with minimal instances of overlap or gaps in the conversation [8, 31]. To enable a smooth and coherent conversation, participants have to engage in turn-taking behaviors when they wish to communicate. Typical cues for initiating a turn include fillers such as "uh" [25]. However, it is a natural phenomenon in natural conversation that the participants speak simultaneously for several reasons [8]. Consequently, turn management can be considered an essential social skill [32], as it enables seamless communication between all participants [2]. It is, therefore, crucial to be able to handle different disruptions [14], considering the context of the conversation [25].

Differentiating overlap, interruption, and back-channeling. When conversation is not effectively coordinated [7], individuals tend to change roles, alternating between speaker and listener. Such behavior results in the occurrence of silences, overlaps, and interruptions [31]. According to Yang et al., [33], nearly 40 percent of natural conversations contain overlap, though not all instances of overlap indicate an interruption [6, 19]. In this context, acoustic features happen to be the key to differentiating overlap and interruption [33]. An interruption occurs when the listener anticipates the speaker's further content and takes over before the speaker has finished, thereby violating the speaker's right to continue speaking [8, 19, 27, 31]. Generally, interruptions can be divided into two categories: "cooperative" and "competitive" [27, 33]. Cooperative interruptions indicate agreement with the speaker or offer assistance, whereas competitive interruptions aim to claim the floor by expressing disagreement or proposing a change of topic [8, 31, 33]. Back-channeling refers to short expressions used by listeners to express their attention and understanding [19]. Typically, the context of the human input is essential. One input from the user can be interpreted differently depending on the situation [34]. For example, the word "uh" can indicate understanding and acknowledgment when pronounced as a short uttering, but it can signal confusion when spoken with a raised pitch at the end. Depending on the interpretation, different actions are required of the interlocutor. Therefore it is crucial to differentiate between overlaps that signal interruptions and those that simply express consent [32].

Interruption in human-IVA interaction. Interruptions are a critical component of natural conversation, and accounting for them is essential to improving the flow and realism of human-IVA interactions. Prior research has shown that enhancing turn-taking behavior can improve communication quality and user engagement [32]. Yang et al. [32, 34] emphasize that user satisfaction increases when an IVA can manage or respond to interruptions effectively, whether initiated by the user or by the system itself. In contrast to traditional robots that rely on pause thresholds to determine turn boundaries, IVAs often suffer from abrupt interruptions or delayed responses [31]. Although natural conversations allow for short overlaps and gaps, successful interaction depends on fluid transitions and the ability to anticipate turn endings [21]. In light of the growing use of speech-based interfaces, IVAs should be equipped with robust interruption management capabilities, including continuous turn-taking—where the microphone remains active and eliminates the need for a repeated wake-up word [21]. However, this requires handling background noise and distinguishing between relevant user input and unrelated side conversations. Our work contributes to this objective by implementing and evaluating real-time verbal interruption in a VR-based IVA. Users were able to naturally interrupt the agent using either short commands or full utterances, creating smoother turn transitions and a more intuitive, human-like interaction flow than what is achievable with pause-based or fixed-turn systems.

3 Methods

This section outlines our methods for conducting the study including the application, the procedure and our methods. We employed a within-subject user study design where participants interacted with

both interruptible IVA and non-interruptible IVA, with condition order counterbalanced. Their task was to answer a list of questions by asking the IVA for the correct answer.

3.1 Conditions

VR Environment: As shown in Figure 1, the application contains a scene in a living-room-like atmosphere, wherein the participant and the IVA are seated across from one another at a table and can engage in communication. The table is furnished with a button to initiate the conversation. A button to end the conversation is visible and can be pressed as soon as the conversation is started. A microphone indicator, colored red to denote that it is currently recording, and white if it is turned off, serves to visually signal the state of the microphone. Participants held their conversations with the IVAs in the same VR environment across different conditions.

Interruptible Condition: When the interruption feature is enabled, the IVA is engaged in speech and the participant wishes to interrupt, they may do so by directly speaking to the IVA. There are cases to distinguish because one has to differentiate between a real interruption and back-channeling [34]. If the user seeks to verbally interrupt, they can use words like “excuse me”, “stop” or simply vocalize their interruption. The system distinguishes between two types of interruption. If the participant only uses the interruption words classified by us beforehand, the IVA immediately stops speaking and switches to listening mode without any new feedback. These words were assessed empirically through pre-testing and include “stop”, “excuse me”, “hold on”, “wait”, “pause”, “no”, and “but”. If multiple words are spoken (more than three), e.g., if a new question is asked, the IVA also interrupts its output, processes the interruption, and responds to the user’s new input directly. As the microphone is always on and the system is sensitive to the noises, it is important that the system does not misinterpret the IVA’s voice as an interruption or other background noise. To solve this issue, we implemented a hardware-based solution. The volume indicator of the Meta Quest 3 was always set to 40 to 50 %. This way, the participants could hear the IVA clearly without the occurrence of microphone loopback.

Non-interruptible Condition: In the condition without the interruption functionality, the user has to wait until the IVA is finished. When the microphone indicator turns red again, the participant is able to input a new question.

3.2 System Implementations

Implementation of the IVA: For integrating the IVAs, we used the Intelligent Virtual Humans SDK [23]. The SDK already allows simple verbal communication with the IVA using the Google Speech-To-Text (STT) API [10], but it mutes the participant’s microphone when the IVA is currently speaking. To better control the interruption settings, we used the dictation feature from Windows [30] instead. Using this, we could customize when the IVA starts listening for new input, and also integrate timeout limits. In our case, the user can take a break of up to 3 seconds, after which the recognition times out and the input will be processed. After speaking, the transcribed words are sent to OpenAI’s GPT-4o mini, which provides an answer to the participant’s question. Imitating a conversation with

a human, the IVA returns the answer via Text-To-Speech (TTS) in a man’s voice, again provided by OpenAI. In order to denote the status of the application, we added a thought bubble (see Figure 1) that is displayed above the IVA while the system processes the user input.

Hardware and Software. : Our application was created using the Unity game engine [29] (version 2022.3.50f1). To display our application and to interact with it in VR, we used a Meta Quest 3 head-mounted display (HMD) which we connected using Quest Link to a Windows computer with an Intel Core i9-9900K CPU, 32 GB RAM and an NVIDIA GeForce RTX 2080 Ti graphics card.



Figure 1: A thought bubble indicates that the IVA is processing the user’s input

3.3 Tasks & Participants

Tasks: A list of questions is provided to direct the conversation. To begin with, a control panel is used to determine which condition (with or without interruption) and - to prevent the repetition of questions - which tasklist (A or B, see Appendix A) is used. The tasklist consists of four questions with single-choice answers. The participant is asked to answer these questions and select the correct answer in the shortest possible time.

Participants: 30 users took part in the study. Of these, 23 were male and 7 female. The mean age was 25.7 years (SD=4.557). Participants were acquired through personal connections and through the institute’s online experimental participation system.

3.4 Measurements

We employed a mixed-method approach in this exploratory study. That is, we used both quantitative and qualitative methods to capture most of the user’s experience with the application.

User Experience Questionnaire: For quantitative methods, we selected the User Experience Questionnaire (UEQ) [18], which provides insight mainly about the usability of the respective systems

and how that affects user experience. It consists of 6 subscales with 26 items: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty. The items are scaled from -3 to +3.

Godspeed: A noteworthy questionnaire for evaluating embodied IVAs is the Godspeed questionnaire (GQ) [4]. Aneja et al. [1] adapted this questionnaire in their study to be used for IVAs. Likewise, we used an abridged version of the Godspeed Questionnaire that was rated on a 5-point Likert scale, only covering the three subscales anthropomorphism, likeability and perceived intelligence. It was employed in German. We chose it as it covers important questions about the perception of the systems, which are important in assessing the effect of interruption on human-IVA interactions. Since all subscales can be evaluated as stand-alone questionnaires [3], we chose the three ones most important for our study.

Qualitative: For qualitative methods, we chose open-ended questions, as they allowed us to gather complimentary information about the participants' experience that standard questionnaires could not capture, e.g., surprises or suggestions for future research. The interview data were reviewed manually to identify key themes and insights that provided contextual understanding for the quantitative findings without following a formal evaluation method.

Participants were asked to answer both UEQ and Godspeed questionnaires after each of the two test runs and answer the open-ended question after completing the experiment.

3.5 Study procedure

In accordance with the ethical guidelines of the ethics commission of our university, a standard user study questionnaire was completed, which indicated that formal ethical approval was not required. Nevertheless, we engaged in a voluntary discussion with the commission to identify possible risks and mitigation strategies, especially related to the usage of AI and IVAs. Before the study commenced, participants were informed that the application incorporates AI-based services, including LLMs, whose outputs may not be entirely predictable. Although content filters are employed to block harmful material such as hate speech, violence, and misinformation, these measures are not infallible, and the LLMs are not customized to individual user sensitivities.

To acknowledge this, participants were asked to read and sign an informed consent form containing general information about the study, its procedures, data collection and management process, and AI usage risks and limitations. Then they generated a code that was used to pseudonymize their data. This ensured privacy while giving the participants the option to demand the deletion of the collected data. The experimenter then summarized the procedure and participants could clarify unanswered questions. The participants were told beforehand that the IVA they were interacting with could be interrupted in one condition, but not the other. This way, they knew that interrupting the IVA was an option, but not a direct requirement.

The participant then put on the HMD and controllers and was given a list of single-choice questions, e.g., tasklist A in the virtual environment to complete during the first experimental condition. After answering all questions, the participant took off the headset and answered the GQ [4] and the UEQ [18]. Then, they answered

open-ended questions about their experience during the study (see Appendix B). This was repeated for the second experimental condition with the other task list of questions (see Appendix A). They then completed an online survey on demographics (gender and age) and their prior experience with VR and chatbots.

4 Results

An alpha level of $\alpha = .05$ was utilized to determine significant results. The statistical software JASP was used for the evaluation. The results can be seen in Table 1

Table 1: Comparison of UEQ and GQ scale scores between conditions with and without interruption

Scale	w/o interr.		with interr.		p	Cohen's d
	M	SD	M	SD		
UEQ: Attractiveness	0.60	1.05	1.13	0.86	0.016	-0.410
UEQ: Perspicuity	1.57	0.90	1.68	0.99	0.348	-0.072
UEQ: Efficiency	0.16	1.28	0.87	1.13	0.011	-0.444
UEQ: Dependability	1.00	0.80	1.08	0.90	0.367	-0.063
UEQ: Stimulation	0.22	1.17	1.00	0.91	0.005	-0.499
UEQ: Novelty	0.43	1.02	0.86	0.91	0.061	-0.290
GQ: Anthropomorphism	2.380	0.758	2.767	0.743	0.066	-0.283
GQ: Likeability	3.667	0.802	3.960	0.648	0.078	-0.266
GQ: Perceived Intelligence	3.787	0.495	3.927	0.569	0.170	-0.177

4.1 User Experience Questionnaire (UEQ)

We computed the mean scores of each subscale of UEQ, see Figure 2. Hinderks et al. provide an official Excel sheet on their website for the evaluation of the subscales, as well as the option to compare the results to a benchmark dataset [12]. For the condition without interruption, the results indicate low Attractiveness, Efficiency and Stimulation score as compared with the benchmark dataset of the questionnaire. Further, the scores for Dependability and Novelty were considered as below average. For the condition with interruptions, these scores were higher but still considered as below average. After this, we used directed paired samples t-tests on the subscales to determine if the means differ significantly. Figure 2 allows visualization of the differences between those two conditions. A significant difference was identified in the subscales of Attractiveness, Efficiency, and Stimulation, indicating that **the interruptible condition was perceived as significantly more attractive, more efficient and more stimulating than the non-interruptible condition.**

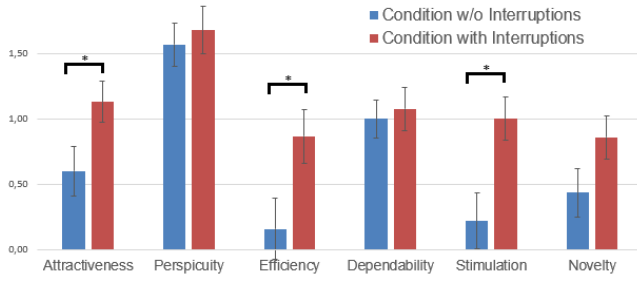


Figure 2: Comparison of UEQ means and the corresponding standard errors. Significant differences marked with an asterisk.

4.2 Godspeed Questionnaire

Likewise, the participant’s impression of the IVA was assessed on a 5-point Likert scale using the three chosen subscales of the Godspeed Questionnaire: Anthropomorphism, Likeability and Perceived Intelligence, see Figure 3.

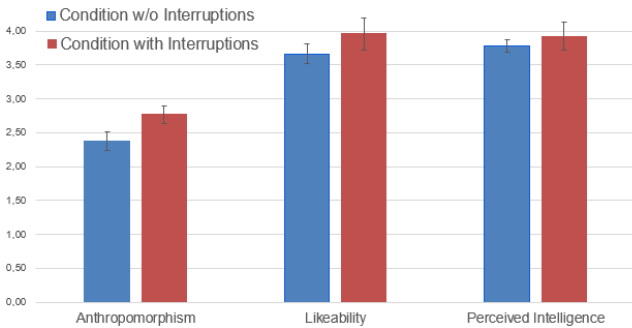


Figure 3: Comparison of GQ scale means and the corresponding standard errors

Shapiro-Wilk tests suggested no deviation from normality for any of the tested value pairs. Using directed, paired samples t-tests on the subscales, no significant results could be found, see Table 1. **This indicates that the possibility to interrupt the IVA did not affect the how users perceived the human-like attributes of the IVA in terms of anthropomorphism, likeability, or perceived intelligence.**

5 Discussion

In this section, we discuss our hypothesis according to the quantitative and qualitative results, and present the limitations and future research directions.

5.1 Interruptibility and User Experiences (H1)

Our statistical analysis revealed significant differences between conditions for the *Attractiveness*, *Efficiency*, and *Stimulation* subscales in the UEQ, partly supporting **H1**. The user experience is improved with the ability to interrupt the IVA. In contrast, several participants gave the feedback that the non-interruptible condition felt

unnatural or frustrating. About 40% of participants noted that the IVA gave overly long responses, with one stating the answers were “way too long for a natural conversation,” and another describing the interaction as a “waiting game for something that does not need further explanation.” These extended responses suggested lower perceived sense of engagement and frustrations with the disrupted conversational flow, which likely led to the lower *Stimulation* scores in the non-interruptible condition. As participants often anticipated a concise answer to a simple question (N=4), some participants even adapted their questioning strategies, either by rephrasing questions or providing list-like prompts to encourage more concise responses. One participant explicitly expressed the wish for a way to prompt the IVA to shorten its answers. These comments align with the significant difference observed in the *Efficiency* subscale: interruptions allowed participants to skip irrelevant or redundant information, making the interaction feel more purposeful and less tedious. Together, the quantitative results and qualitative feedback suggest that the ability to manage turn-taking, particularly through interruptions, is not only functionally helpful but also critical for supporting efficient and stimulating communication with IVAs. These findings are consistent with previous literature describing the management of turn-taking cues as essential [32] and important for seamless communication [2].

It is worth noting that the effect sizes for the subscales with significant differences ranged between 0.4 and 0.5, indicating medium effects. This suggests that the ability to interrupt had a moderate positive impact on user experience, enhancing the overall impression of the interaction, as well as engagement and perceived efficiency. However, the absolute scores across subscales remained relatively low, highlighting room for improvement in the communication between the IVA and the user. For example, participants mentioned that after interrupting the IVA, there was still a delay until it stopped talking, which did not seem like how humans naturally behaved (N=3). Since the IVA took time to process input and the conversation was more interactive, the waiting time for the feedback was sometimes even longer (N=1). In prior work, a high delay in receiving an answer from an AI also led to a deteriorated user experience [17], and frustration when needing timely assistance [5]. Therefore, we suggest that future systems not only support interruption, but also respond to it more promptly and fluidly, minimizing latency and aligning more closely with human conversational norms. Improving response timing and responsiveness may help create a more natural dialogue flow and increase users’ overall satisfaction with the interaction.

5.2 Interruptibility and Naturalness (H2)

Regarding the results of the Godspeed questionnaire, no significant differences were found between the two conditions across the subscales of *Anthropomorphism*, *Likeability*, or *Perceived Intelligence*. As a result, **H2** could not be supported. While the overall mean scores were slightly higher in the interruptible condition, the effect may not have been strong enough to reach statistical significance with the given sample size (N=30). Nonetheless, many participants described the interruptible IVA as more fluid (N=1), balanced (N=1), and natural (N=12), noting that the ability to interrupt aligned better

with how people typically manage turn-taking in human conversations. Some also mentioned that the system responded promptly to their interruptions, reinforcing the perception of natural interaction ($N=1$). These responses suggest that the interruptibility feature contributed positively to perceived conversational quality, even if this was not reflected strongly in the Godspeed scores.

At the same time, some participants voiced concerns that the interaction still lacked key features of human conversation. For instance, the IVA never reacted to being interrupted, which some found unrealistic—highlighting that real people typically show at least minimal acknowledgment or resistance when interrupted ($N=1$). Participants described their own interruption strategies using phrases like “stop” ($N=3$), “wait” ($N=1$), “thank you, next question” ($N=1$), “alright” ($N=1$), or by simply posing the next question directly ($N=2$). These varying approaches underscore that what is considered a “natural” way to interrupt a conversational partner can differ significantly among users. Such diversity suggests that future systems could benefit from recognizing a broader range of interruption cues and responding in more socially aware ways.

Interestingly, several participants noted that their behavior with the IVA did not mirror how they would interact with a real person. One participant commented that the exchange felt like talking to a Google search history rather than a conversational partner. Another pointed out that the IVA never questioned the interruption and simply moved on to the next answer, which made the dialogue feel transactional. Furthermore, another participant mentioned deliberately using the word “stop,” something they would not typically say in a natural conversation, but did so “when talking to a machine.” In contrast, one participant said they remained polite and avoided being rude, showing signs of anthropomorphizing the IVA. These mixed impressions illustrate that while participants appreciated the added control that came with interruptibility, the feature alone was not enough to create a fully human-like interaction.

Taken together, the qualitative responses suggest that while the ability to interrupt is a valued and functionally useful feature, it can also challenge users’ expectations of social reciprocity. For interruptions to truly enhance perceived naturalness, future designs should consider both responsiveness and the contextual appropriateness of the IVA’s behavior after being interrupted.

5.3 Future Work & Limitations

Non-verbal Interruptions: Participants offered varied opinions on possible non-verbal interruption strategies. Some did not see a need for them, stating that verbal interruptions felt more intuitive ($N=3$). Others felt that non-verbal cues could be more polite in certain contexts ($N=2$). Nearly half of the participants suggested using hand gestures such as raising a hand, waving, or showing the palm in a stop motion. A few mentioned head movements or facial expressions ($N=3$), and some preferred a simple interruption button ($N=3$). One participant also suggested that the IVA could detect when users were distracted or no longer actively listening, and stop speaking automatically. Another participant recommended offering users the option to customize their preferred interruption method. Prior research shows that the process of turn-taking itself is inherently multimodal, with gaze, gestures, and body language playing critical

roles in coordinating turn-taking interactions [15]. Further, communication strategies can differ between different people, between languages [24], and between genders [13]. Since we mostly tested young males from Germany in our study, this poses a limitation. In future studies, the communication with an IVA should be explored with a more balanced sample, and individual preferences should be accounted for. These suggestions highlight the value of exploring multimodal input strategies that better align with users’ diverse expectations and communication habits [15].

User Intent Estimation via Speech: Our current implementation balances responsiveness with robustness by limiting interruptions to certain stop words or utterances of three or more words. However, users’ backchanneling attempts, particularly those longer than a few words, can unintentionally trigger an interruption. Future systems should aim to more accurately detect the user’s intent to interrupt. One possible approach would be training a local AI model to differentiate between backchanneling and genuine interruption attempts based on both content and context. Integrating non-verbal cues such as facial expressions, eye gaze, or gestures could further improve interruption timing and help the IVA better interpret users’ communicative intent. This would not only reduce false positives but also enhance the overall fluidity and efficiency of the interaction. In recent work, the performance of a multimodal transformer-based Voice Activity Projection model for turn-taking incorporating non-verbal features has been evaluated, and it showed improvements compared to an unimodal model [24]. Nevertheless, its incorporation into real virtual agents has not been evaluated yet, and, therefore, research on this topic remains current.

Improvement on Behavioral Fidelity: Several design limitations of our IVA may have affected users’ perception of naturalness. These include its visual appearance, gender representation, and the lack of synchronized or expressive non-verbal behavior. Future iterations could enhance the agent’s realism by incorporating subtle bodily and facial animations—such as nodding, maintaining eye contact, or showing “thinking” gestures—instead of relying on static or text-based visual indicators like emojis. Further, while our study task to answer quiz questions provided a framework for shaping the conversation, future studies should test this with more natural, open-ended conversations to enhance external validity of our results.

Design Implications and Ethical Considerations: Looking ahead, comparative studies should evaluate different interruption modalities (e.g. verbal, gestural, and multimodal) to better understand how they affect user experience and perceived naturalness. Interestingly, while physical buttons may seem unnatural in human-human interaction, some participants expressed a preference for such options when engaging with a virtual agent in VR. This suggests that users still view the IVA as distinct from a real person, and may therefore accept less human-like controls in exchange for functional benefits. These findings support the idea that, while IVAs strive to emulate human behavior, it remains important to clearly communicate their artificial nature. As discussed in recent literature [26], transparent design choices help maintain ethical boundaries and prevent unintentional deception. In our study, this could be related to using stop

words or visible UI elements. Rather than strictly mimicking human norms, future work should also explore hybrid strategies that blend human-like interaction with practical affordances tailored to human-AI communication.

6 Conclusion

We implemented and evaluated the ability to verbally interrupt an IVA in VR through a user study with 30 participants. Results showed that interruptibility significantly improved user experience, particularly in terms of perceived attractiveness, efficiency, and stimulation. Participants described the interaction as more natural and fluid when interruptions were possible, though views varied, some found verbal interruptions intuitive and human-like, while others saw them as impolite. These mixed responses point to the importance of personalization in IVA design. While beneficial, interruptibility introduces technical challenges such as distinguishing intentional interruptions from back-channeling. We outline directions for future research, including more sophisticated detection methods and the integration of non-verbal cues. Overall, our findings highlight the value of supporting interruptions to enhance conversational flow and responsiveness in VR-based human-agent interactions.

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References

- [1] Deepali Aneja, Daniel McDuff, and Mary Czerwinski. 2020. Conversational Error Analysis in Human-Agent Interaction. In *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents* (Virtual Event, Scotland, UK) (IVA '20). Association for Computing Machinery, New York, NY, USA, Article 3, 8 pages. doi:10.1145/3383652.3423901
- [2] Young-Ho Bae and Casey C. Bennett. 2023. Real-Time Multimodal Turn-taking Prediction to Enhance Cooperative Dialogue during Human-Agent Interaction. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. 2037–2044. doi:10.1109/RO-MAN57019.2023.10309569
- [3] Christoph Bartneck. 2023. Godspeed questionnaire series: Translations and usage. In *International handbook of behavioral health assessment*. Springer, 1–35.
- [4] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International journal of social robotics* 1 (2009), 71–81.
- [5] Mouad Boumia et al. 2025. A STUDY ON CHATBOT SERVICE FAILURES IN E-COMMERCE AND IMPROVEMENT STRATEGIES OF AMAZON. *Policy Journal of Social Science Review* 3, 5 (2025), 313–331.
- [6] Angelo Cafaro, Brian Ravenet, and Catherine Pelachaud. 2019. Exploiting evolutionary algorithms to model nonverbal reactions to conversational interruptions in user-agent interactions. *IEEE Transactions on Affective Computing* 13, 1 (2019), 485–495.
- [7] Aleksandra Cerekovic, Oya Aran, and Daniel Gatica-Perez. 2017. Rapport with Virtual Agents: What Do Human Social Cues and Personality Explain? *IEEE Transactions on Affective Computing* 8, 3 (2017), 382–395. doi:10.1109/TAFFC.2016.2545650
- [8] Ronald Cumbal, Reshma Kantharaju, Maike Paetzel-Prüsmann, and James Kennedy. 2024. Let me finish first-The effect of interruption-handling strategy on the perceived personality of a social agent. In *24th ACM International Conference on Intelligent Virtual Agents*.
- [9] Yang Gao, Yangbin Dai, Guangtao Zhang, Honglei Guo, Fariba Mostajeran, Bing Zheng, and Tao Yu. 2025. Trust in Virtual Agents: Exploring the Role of Stylization and Voice. *IEEE Transactions on Visualization and Computer Graphics* (2025).
- [10] Google Cloud EMEA Limited. 2023. *Google Speech-To-Text API*. <https://cloud.google.com/speech-to-text/docs/reference/rest/v1/speech/recognize> Google API.
- [11] Shirin Hajahmadi, Pasquale Cascarano, Fariba Mostajeran, Kevin Heuer, Anton Lux, Gil Otis Mends-Cole, Frank Steinicke, and Gustavo Marfia. 2025. Investigating the Impact of Voice-only and Embodied Conversational Virtual Agents on Mixed Reality Puzzle Solving. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 602–612.
- [12] Andreas Hinderks, Martin Schrepp, and Jörg Thomaschewski. 2024. User Experience Questionnaire. <https://www.ueq-online.org/>
- [13] Janet Holmes and Nick Wilson. 2022. *An introduction to sociolinguistics*. Routledge. doi:10.4324/9780367821852
- [14] Kathrin Janowski and Elisabeth André. 2019. What If I Speak Now? A Decision-Theoretic Approach to Personality-Based Turn-Taking. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems* (Montreal QC, Canada) (AAMAS '19). International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 1051–1059.
- [15] Kobin H. Kendrick, Judith Holler, and Stephen C. Levinson. 2023. Turn-taking in human face-to-face interaction is multimodal: gaze direction and manual gestures aid the coordination of turn transitions. *Philosophical Transactions of the Royal Society B* 378 (2023). <https://api.semanticscholar.org/CorpusID:257354004>
- [16] Lucie Kruse, Fariba Mostajeran, and Frank Steinicke. 2023. The Influence of Virtual Agent Visibility in Virtual Reality Cognitive Training. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction*. 1–9.
- [17] Lucie Kruse, Sebastian Rings, and Frank Steinicke. 2025. My Focus was on the Game: Investigating the Use of AI Assistants in a Virtual Reality Exergame for Older Adults. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [18] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and Evaluation of a User Experience Questionnaire. *USAB 2008* 5298, 63–76. doi:10.1007/978-3-540-89350-9_6
- [19] Meng-Chen Lee and Zhigang Deng. 2024. Online Multimodal End-of-Turn Prediction for Three-party Conversations. In *Proceedings of the 26th International Conference on Multimodal Interaction* (San Jose, Costa Rica) (ICMI '24). Association for Computing Machinery, New York, NY, USA, 57–65. doi:10.1145/3678957.3685742
- [20] Fu Li, Fariba Mostajeran, Frank Steinicke, and Matthew HEM Browning. 2024. Extended reality in environmental neuroscience research. In *Environmental Neuroscience*. Springer, 455–490.
- [21] Yohan Moon, Yeri Jeong, and Eugene Seo. 2019. I'm listening: the effect of cue difference to elicit user's continuous turn-taking with A.I. agent in TV. In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers* (London, United Kingdom) (UbiComp/ISWC '19 Adjunct). Association for Computing Machinery, New York, NY, USA, 153–156. doi:10.1145/3341162.3343848
- [22] Fariba Mostajeran, Nadia Burke, Nazife Ertugrul, Kilian Hildebrandt, Joshua Matov, Noémie Tapie, Wilhelm Gottlieb Zittel, Pia Reisewitz, and Frank Steinicke. 2022. Anthropomorphism of virtual agents and human cognitive performance in augmented reality. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 329–332.
- [23] Fariba Mostajeran, Ke Li, Sebastian Rings, Lucie Kruse, Erik Wolf, Susanne Schmidt, Michael Arz, Joan Llobera, Pierre Nagorny, Caecilia Charbonnier, Hannes Fassold, Xenxo Alvarez, Tavares André, Nuno Santos, João Orvalho, Sergi Fernández, and Frank Steinicke. [n. d.]. A Toolkit for Creating Intelligent Virtual Humans in Extended Reality. <https://www.charbonnier.ch/pdf/publications/118.pdf> Accessed: 17 March 2025.
- [24] Kazuyo Onishi, Hiroki Tanaka, and Satoshi Nakamura. 2024. Multimodal voice activity projection for turn-taking and effects on speaker adaptation. *IEICE Transactions on Information and Systems* (2024).
- [25] Pourya Shahverdi, Alexander Tyshka, Madeline Trombly, and Wing-Yue Geoffrey Louie. 2022. Learning Turn-Taking Behavior from Human Demonstrations for Social Human-Robot Interactions. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 7643–7649. doi:10.1109/IROS47612.2022.9981243
- [26] Judith Simon, Indra Spieker Döhmman, and Ulrike von Luxburg. 2024. Generative AI - Beyond Euphoria and Simple Solutions. *Discussions No. 34* (2024). doi:10.26164/leopoldina_03_01245
- [27] Gabriel Skantze. 2021. Turn-taking in Conversational Systems and Human-Robot Interaction: A Review. *Computer Speech & Language* 67 (2021), 101178. doi:10.1016/j.csl.2020.101178
- [28] Mel Slater, Domna Banakou, Alejandro Beacco, Jaime Gallego, Francisco Macia-Varela, and Ramon Oliva. 2022. A separate reality: An update on place illusion and plausibility in virtual reality. *Frontiers in virtual reality* 3 (2022), 914392.
- [29] Unity Technologies. 2023. *Unity*. <https://unity.com/> Game development platform.

- [30] Unity Technologies. 2025. *Unity DictationRecognizer*. <https://docs.unity3d.com/2022.3/Documentation/ScriptReference/Windows.Speech.DictationRecognizer.html> Unity component.
- [31] Liu Yang. 2021. What If I Interrupt You. In *Proceedings of the 2021 International Conference on Multimodal Interaction* (Montréal, QC, Canada) (ICMI '21). Association for Computing Machinery, New York, NY, USA, 827–831. doi:10.1145/3462244.3481278
- [32] Liu Yang, Catherine Achard, and Catherine Pelachaud. 2021. Interruptions in Human-Agent Interaction. In *Proceedings of the 21st ACM International Conference on Intelligent Virtual Agents* (Virtual Event, Japan) (IVA '21). Association for Computing Machinery, New York, NY, USA, 206–208. doi:10.1145/3472306.3478340
- [33] Liu Yang, Catherine Achard, and Catherine Pelachaud. 2022. Multimodal classification of interruptions in humans' interaction. In *Proceedings of the 2022 International Conference on Multimodal Interaction* (Bengaluru, India) (ICMI '22). Association for Computing Machinery, New York, NY, USA, 597–604. doi:10.1145/3536221.3556604
- [34] Liu Yang, Catherine Achard, and Catherine Pelachaud. 2023. Now or When? Interruption timing prediction in dyadic interaction. In *Proceedings of the 23rd ACM International Conference on Intelligent Virtual Agents* (Würzburg, Germany) (IVA '23). Association for Computing Machinery, New York, NY, USA, Article 44, 4 pages. doi:10.1145/3570945.3607293

A Tasklists used in the study

Tasklist A:

- (1) What is Wikipedia?
 - Encyclopedia
 - Dictionary
 - Database
 - Anthology
- (2) What is a Portuguese man o' war?
 - A jellyfish with deadly stingers
 - A famous pirate from Portugal
 - A type of battleship used in ancient Portugal
 - A siphonophore, a marine organism
- (3) Why did October 1582 only have 3 weeks?
 - The Augustine calendar was abandoned, removing 10 days
 - Earth's rotation slowed, shortening the month
 - A change in the moon's cycle shortened October
 - The Gregorian calendar skipped 10 days to fix seasons
- (4) Who created the sandwich?
 - Anita Baker, the famous cook
 - John Montagu, the 4th Earl of Sandwich
 - Heston Blumenthal, using liquid nitrogen for an "instant sandwich"
 - Marie Antoinette, is said to have invented it at a royal feast

Tasklist B:

- (1) Botanically, which of the following is classified as a berry?
 - Banana
 - Strawberry
 - Raspberry
 - Blackberry
- (2) Which city is also called "ice city"?
 - Reykjavik (Iceland), because of the cold climate
 - Moscow, Russia, due to its snowy winters
 - Harbin, China, because of its famous ice festival
 - Oslo, Norway, because of its icy fjords
- (3) Which is the highest-grossing film (Okt. 2023)?
 - Titanic
 - Avatar
 - Avengers: Endgame

- Star Wars: The Force Awakens

- (4) Where did the term "spam" originate in relation to spam emails?

- Monty Python's Flying Circus sketch
- A marketing campaign for canned meat
- An abbreviation for "Special Promotional Advertisement Mail"
- The first major email service provider, SpamMail

B Open-Ended Questions

- (1) Did the conversation with the agent feel natural? Why?
- (2) Do you think, in case you tried to interrupt the agent, the way the agent reacted to your interruptions was adequate?
- (3) If you were able to interrupt the agent, do you think the ways of interruption (words/phrases that were used) are intuitive? Which ways of interruption did you expect?
- (4) What features were well implemented and which can be improved?
- (5) For future research, do you think you would use non-verbal interruptions? Which non-verbal interruptions would be useful?